

Laporan Akhir
Data Science Deteksi Umur Dengan Wajah



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Spesifikasi Modul Perangkat Lunak

1. Modul Utilitas (Performa & Stabilitas)

- **Collections.deque:** Berfungsi sebagai penyimpan riwayat prediksi usia dan gender dalam jangka pendek menggunakan sliding window. Modul ini mengumpulkan hasil estimasi dari beberapa frame terakhir kemudian menghitung rata-ratanya, sehingga nilai yang ditampilkan lebih stabil dan tidak berubah-ubah secara drastis antar frame.
- **NumPy:** Menangani komputasi numerik dan matematis pada data array, seperti menghitung nilai rata-rata (mean) dari riwayat estimasi usia yang tersimpan didalam deque, serta membantu pemrosesan koordinat bounding box wajah.
- **Os:** Mengelola operasi sistem berkaitan dengan file dan direktori, seperti memisah nama file dari ekstensinya saat menyimpan hasil deteksi gambar kedalam penyimpanan lokal.

2. Modul Utama (Pemrosesan Citra & AI)

- **OpenCV (cv2):** Menangani seluruh proses pemrosesan citra secara real-time, mulai dari mengakses hardware Webcam, membaca frame video, melakukan flip gambar, menggambar teks label usia dan gender, hingga menampilkan hasil ke layout pengguna
- **InsightFace (buffalo_1):** Merupakan mesin utama kecerdasan buatan dalam sistem ini. Model buffalo_1 menerima frame gambar, mendeteksi lokasi wajah menggunakan modul detection, kemudian memproses potongan wajah tersebut melalui modul genderage yang menggunakan arsitektur deep learning berbasis ONNX untuk menghasilkan prediksi usia dan jenis kelamin secara bersamaan.

Source Code

```
import cv2
import numpy as np
from insightface.app import FaceAnalysis
import os
from collections import deque

# -----
# Konfigurasi
# -----

AGE_RANGES = [
    (0, 2, "0-2 yr"),
    (3, 6, "3-6 yr"),
    (7, 12, "7-12 yr"),
    (13, 17, "13-17 yr"),
    (18, 24, "18-24 yr"),
    (25, 32, "25-32 yr"),
    (33, 40, "33-40 yr"),
    (41, 50, "41-50 yr"),
    (51, 60, "51-60 yr"),
    (61, 99, "61+ yr"),
]

# Koreksi offset: InsightFace cenderung overestimate ~12-15 tahun
# untuk wajah Asia Tenggara. Sesuaikan angka ini jika perlu.
AGE_OFFSET = -13

def age_to_range(age: int) -> str:
    for low, high, label in AGE_RANGES:
        if low <= age <= high:
            return label
    return f"{age} yr"

def load_model():
    app = FaceAnalysis(
        name="buffalo_l",
        allowed_modules=["detection", "genderage"],
        providers=["CPUExecutionProvider"]
    )
```

```

app.prepare(ctx_id=0, det_size=(320, 320), det_thresh=0.35)
return app

# -----
# Smoothing tracker
# -----

class FaceTracker:
    def __init__(self, window=8):
        self.window = window
        self.age_history = {}
        self.gender_history = {}

    def _face_key(self, bbox):
        x1, y1, x2, y2 = bbox
        cx = int((x1 + x2) / 2 / 50)
        cy = int((y1 + y2) / 2 / 50)
        return (cx, cy)

    def update(self, face):
        key = self._face_key([int(v) for v in face.bbox])
        if key not in self.age_history:
            self.age_history[key] = deque(maxlen=self.window)
            self.gender_history[key] = deque(maxlen=self.window)
        self.age_history[key].append(int(face.age))
        self.gender_history[key].append(int(face.gender))

    def get_smoothed(self, face):
        key = self._face_key([int(v) for v in face.bbox])
        if key in self.age_history and len(self.age_history[key])
> 0:
            smooth_age = int(np.mean(self.age_history[key]))
            gender_votes = list(self.gender_history[key])
            smooth_gender = 1 if gender_votes.count(1) >=
len(gender_votes) / 2 else 0
        else:
            smooth_age = int(face.age)
            smooth_gender = int(face.gender)
        return smooth_age, smooth_gender

# -----

```

```

# Deteksi file gambar
# -----

def detect_image(image_path: str, app: FaceAnalysis, save: bool =
True):
    frame = cv2.imread(image_path)
    if frame is None:
        print(f"[ERROR] File tidak ditemukan: {image_path}")
        return

    h, w = frame.shape[:2]
    if w < 300:
        frame = cv2.resize(frame, (640, int(h * 640 / w)))

    faces = app.get(frame)

    if len(faces) == 0:
        print("[INFO] Tidak ada wajah yang terdeteksi.")
        return

    print(f"\n✓ {len(faces)} wajah terdeteksi!\n")
    print(f"{'No':<5} {'Gender':<10} {'Usia Raw':<10} {'Usia
Koreksi':<14} {'Range':<16}")
    print("-" * 55)

    for i, face in enumerate(faces):
        gender      = "Male" if face.gender == 1 else "Female"
        age_raw     = int(face.age)
        age_corrected = max(1, age_raw + AGE_OFFSET)
        age_range   = age_to_range(age_corrected)

        print(f"{i+1:<5} {gender:<10} {age_raw:<10}
{age_corrected:<14} {age_range:<16}")

        x1, y1, x2, y2 = [int(v) for v in face.bbox]
        color = (255, 100, 50) if gender == "Male" else (50, 200,
255)

        cv2.rectangle(frame, (x1, y1), (x2, y2), color, 2)

        label = f"{gender}, {age_range}"
        (tw, th), _ = cv2.getTextSize(label,
cv2.FONT_HERSHEY_DUPLEX, 0.7, 1)

```

```

        cv2.rectangle(frame, (x1, y1 - th - 12), (x1 + tw + 8,
y1), color, -1)
        cv2.putText(frame, label, (x1 + 4, y1 - 6),
                    cv2.FONT_HERSHEY_DUPLEX, 0.7, (255, 255, 255),
1, cv2.LINE_AA)

    if save:
        base, ext = os.path.splitext(image_path)
        output_path = f"{base}_hasil{ext}"
        cv2.imwrite(output_path, frame)
        print(f"\n📁 Hasil disimpan: {output_path}")

    cv2.imshow("InsightFace – Age & Gender Detection", frame)
    cv2.waitKey(0)
    cv2.destroyAllWindows()

# -----
# Deteksi webcam real-time
# -----

def detect_webcam(app: FaceAnalysis):
    cap = cv2.VideoCapture(0)
    if not cap.isOpened():
        print("[ERROR] Webcam tidak bisa dibuka.")
        return

    cap.set(cv2.CAP_PROP_FRAME_WIDTH, 640)
    cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 480)

    tracker = FaceTracker(window=8)
    frame_count = 0
    last_faces = []

    print("📹 Webcam aktif. Tekan 'q' untuk keluar, 's' untuk
screenshot.\n")

    while True:
        ret, frame = cap.read()
        if not ret:
            break

        frame = cv2.flip(frame, 1)

```

```

frame_count += 1

faces = app.get(frame)
if len(faces) > 0:
    last_faces = faces
else:
    faces = last_faces # anti-flicker

for face in faces:
    tracker.update(face)
    smooth_age, smooth_gender = tracker.get_smoothed(face)

    # Terapkan koreksi offset
    corrected_age = max(1, smooth_age + AGE_OFFSET)

    gender = "Male" if smooth_gender == 1 else "Female"
    age_range = age_to_range(corrected_age)

    x1, y1, x2, y2 = [int(v) for v in face.bbox]
    color = (255, 100, 50) if gender == "Male" else (50,
200, 255)

    cv2.rectangle(frame, (x1, y1), (x2, y2), color, 2)

    label = f"{gender}, {age_range}"
    (tw, th), _ = cv2.getTextSize(label,
cv2.FONT_HERSHEY_DUPLEX, 0.65, 1)
    cv2.rectangle(frame, (x1, y1 - th - 12), (x1 + tw + 8,
y1), color, -1)
    cv2.putText(frame, label, (x1 + 4, y1 - 6),
                    cv2.FONT_HERSHEY_DUPLEX, 0.65, (255, 255,
255), 1, cv2.LINE_AA)

    # Raw age untuk debug
    cv2.putText(frame, f"raw: {smooth_age}",
                    (x1, y2 + 18), cv2.FONT_HERSHEY_PLAIN,
1.0, (200, 200, 200), 1)

    cv2.putText(frame, "Press 'q' quit | 's' screenshot",
                    (10, 25), cv2.FONT_HERSHEY_PLAIN, 1.2, (0,
255, 0), 1)

    cv2.imshow("Age Detection Webcam", frame)

```

```

        key = cv2.waitKey(1) & 0xFF
        if key == ord('q'):
            break
        elif key == ord('s'):
            filename = f"screenshot_{frame_count}.jpg"
            cv2.imwrite(filename, frame)
            print(f"📷 Screenshot disimpan: {filename}")

    cap.release()
    cv2.destroyAllWindows()

# -----
# Entry point
# -----
if __name__ == "__main__":
    print("🔄 Loading model InsightFace...")
    app = load_model()
    print("✅ Model siap!\n")

    # Webcam real-time:
    detect_webcam(app)

    # Untuk deteksi dari file gambar, uncomment baris di bawah:
    # detect_image("17_Disti.jpg", app)
    # detect_image("18_TITI.jpeg", app)
    # detect_image("19_Aza.jpeg", app)

```


Output

